Q2: Does roster continuity help explain team success?

We test whether continuity adds information after accounting for prior team strength and player talent.

How continuity is measured

We define **continuity** as the share of rotation minutes played by players who were on that same team one year earlier. Rotation players must log at least 150 minutes in the season. Player-team minutes are reconstructed from game logs, which handles trades correctly. Talent is measured only from current rotation players' prior-season Player Impact Estimate (PIE). We evaluate the models by leaving out one complete season at a time. The roster summaries use the full season, so this is an explanatory test rather than a preseason forecast.

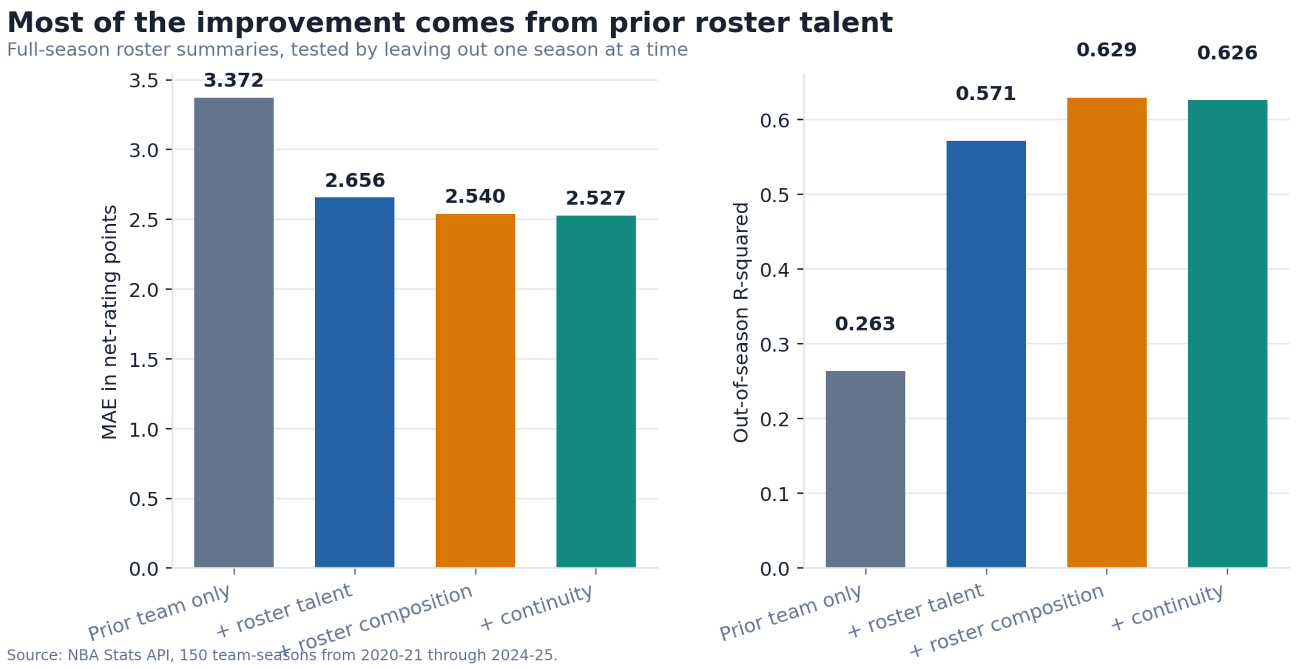
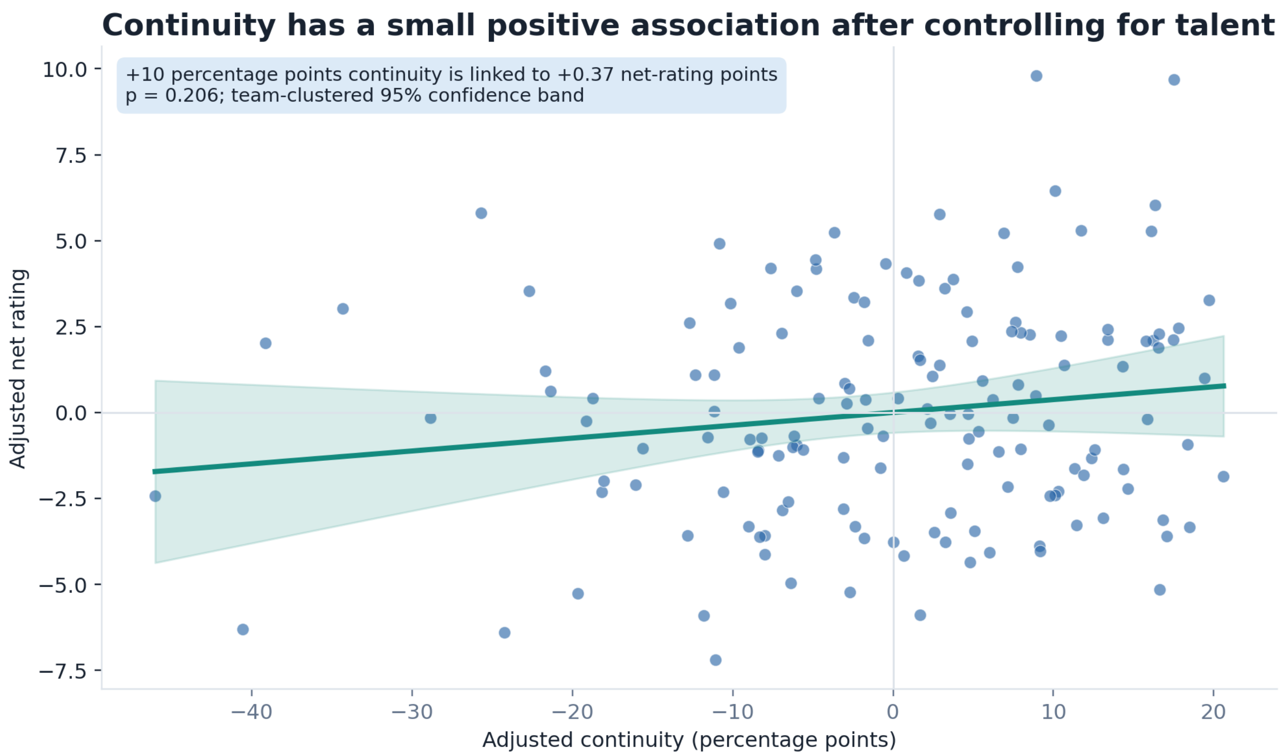


Figure 1. Out-of-season prediction of current team net rating across 150 team-seasons. Roster composition includes age, playing-time concentration, and usage spread. These are full-season summaries. Lower MAE is better, while higher R-squared is better.

Main result: Prior roster talent reduces MAE by 0.72 net-rating points, and composition adds another 0.12. Continuity adds only 0.013 points of improvement (95% team-bootstrap CI: -0.052 to 0.081). R-squared changes from 0.629 to 0.626.

Q2: Continuity after controlling for talent

We remove the part explained by season, prior team strength, and prior player performance.



Controls: season, prior team net rating, prior roster PIE, star PIE, and missing prior data. Association, not causation.

Figure 2. The axes show what remains after controlling for season, prior team net rating, average prior roster PIE, star PIE, and missing prior-season data. The confidence band uses standard errors clustered by team.

An extra 10 percentage points of continuity is associated with +0.37 net-rating points, but the estimate is uncertain ($p = 0.206$). The result suggests a small positive relationship, but it is not statistically significant at the 0.05 level.

Main point: Continuity may help, but most of the predictable difference between teams comes from roster quality and composition.

Q2: Talent and continuity together

We split the teams into higher and lower groups to make the pattern easier to see.

Teams do best when they have both talent and continuity

Average current-season net rating; median splits

Source: NBA Stats API, 2020-21 through 2024-25. Talent uses current-roster players' prior-season PIE.

Figure 3. Average current-season net rating after splitting teams at the sample medians for prior roster talent and continuity. Counts are shown in each cell.

Higher-continuity teams have a better average net rating within both talent groups. However, the talent difference is still clear. Higher-talent teams with lower continuity average +1.8, while lower-talent teams with higher continuity average -1.2. Teams with both higher talent and higher continuity average +3.9.

Q2 conclusion: In this sample, teams do best when they combine prior talent with continuity. Continuity is associated with better results, but talent explains much more of the difference between teams.

Limitations: this is an observational sample of five NBA seasons. End-of-season minutes define the rotation, so injuries and coaching choices affect the measured exposure. The analysis cannot identify a causal effect of continuity or directly observe relationships, leadership, or locker-room dynamics.